ASCE 41-23 Tier 1 Checklists

FIRM:	
PROJECT NAME:	
SEISMICITY LEVEL:	
PROJECT NUMBER:	
COMPLETED BY:	
DATE COMPLETED:	
REVIEWED BY:	
REVIEW DATE:	

Legend: C = Compliant, NC = Noncompliant, N/A = Not Applicable, U = Unknown

© 2023 American Society of Civil Engineers

Project Name _____
Project Number _____

17.8CP Structural Checklist for Building Types S5: Steel Frames with Infill Masonry Shear Walls and Stiff Diaphragms and S5a: Steel Frames with Infill Masonry Shear Walls and Flexible Diaphragms

Table 17-16. Collapse Prevention Structural Checklist for Building Types S5 and S5a.

Status				Evaluation Statement	Tier 2 Reference	Commentary Reference	Comments
Low Seismicity							
Seismic-Force-Resisting System							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	REDUNDANCY: The number of lines of shear walls in each principal direction is greater than or equal to 2.	5.5.1.1	A.3.2.1.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	SHEAR STRESS CHECK: The shear stress in the reinforced masonry shear walls, calculated using the Quick Check procedure of Section 4.4.3.3 is less than 70 lb/in.^{2} (0.48 MPa)	5.5.3.1.1	A.3.2.4.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	SHEAR STRESS CHECK: The shear stress in the unreinforced masonry shear walls, calculated using the Quick Check procedure of Section 4.4.3.3 is less than 30 lb/in.^{2} (0.21 MPa) for clay units and 70 lb/in.^{2} (0.48 MPa) for concrete units. Bays with openings greater than 25% of the wall area shall not be included in A_w of Equation (4-8).	5.5.3.1.1	A.3.2.5.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	INFILL WALL CONNECTIONS: Masonry is in full contact with frame.	5.5.3.5.1 5.5.3.5.3	A.3.2.6.1	
Connections							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	STEEL COLUMNS: The columns in seismic-force-resisting frames are anchored to the building foundation with a minimum of two anchor rods and with the base plates bearing on concrete or a grout pad.	5.7.3.1	A.5.3.1	

Status				Evaluation Statement	Tier 2 Reference	Commentary Reference	Comments
Moderate Seismicity (Complete the Following Items in Addition to the Items for Low Seismicity)							
Seismic-Force-Resisting System							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	INFILL WALL ECCENTRICITY: The centerline of the infill masonry wall is not offset from the centerline of the steel framing by more than 25% of the wall thickness.	5.5.3.5.3	A.3.2.6.5	

Legend: C = Compliant, NC = Noncompliant, N/A = Not Applicable, U = Unknown

Connections						
C	**NC**	**N/A**	**U**	TRANSFER TO INFILL WALLS: Diaphragms are connected for transfer of loads to the infill walls.	5.7.2	A.5.2.1

Status				Evaluation Statement	Tier 2 Reference	Commentary Reference	Comments
High Seismicity (Complete the Following Items in Addition to the Items for Low and Moderate Seismicity)							
Seismic-Force-Resisting System							
C	**NC**	**N/A**	**U**	PROPORTIONS: The height-to-thickness ratio of the unreinforced infill walls at each story is less than 9.	5.5.3.1.2	A.3.2.6.2	
C	**NC**	**N/A**	**U**	CAVITY WALLS: The infill walls are not of cavity construction.	5.5.3.5.2	A.3.2.6.3	
Flexible Diaphragms							
C	**NC**	**N/A**	**U**	CROSSTIES: There are continuous crossties between diaphragm chords.	5.6.1.2	A.4.1.2	
C	**NC**	**N/A**	**U**	STRAIGHT SHEATHING: All straight-sheathed diaphragms have horizontal spans less than 24 ft (7.3 m) and aspect ratios less than 2-to-1 in the direction being considered.	5.6.2	A.4.2.1	
C	**NC**	**N/A**	**U**	DIAGONALLY SHEATHED AND UNBLOCKED DIAPHRAGMS: All diagonally sheathed or unblocked wood structural panel diaphragms have horizontal spans less than 40 ft (12.2 m) and aspect ratios less than or equal to 4-to-1.	5.6.2	A.4.2.2	
C	**NC**	**N/A**	**U**	BLOCKED DIAPHRAGMS: All blocked wood structural panel diaphragms have horizontal spans less than 120 ft (36.5 m) and have aspect ratios less than or equal to 4-to-1.	5.6.2	A.4.2.3	
C	**NC**	**N/A**	**U**	CANTILEVERED WOOD DIAPHRAGMS: All cantilevered wood diaphragms that provide lateral support for concrete or masonry walls consist of wood structural panels and have a maximum cantilever length of 20 ft (6.1 m) if unblocked or 35 ft (10.7 m) if blocked, and a maximum ratio of cantilever length to diaphragm width of 1:2 if unblocked and 1:1 if blocked. In addition, the cantilevered diaphragm has a back-span length equal to or greater than the cantilevered portion.	5.6.2	A.4.2.4	

Legend: C = Compliant, NC = Noncompliant, N/A = Not Applicable, U = Unknown

© 2023 American Society of Civil Engineers

C	NC	N/A	U			
☐	☐	☐	☐	NON-CONCRETE-FILLED DIAPHRAGMS: Bare steel deck diaphragms or steel deck diaphragms with fill other than reinforced structural concrete consist of horizontal spans of less than 120 ft (36.5 m) and have aspect ratios less than 4-to-1.	5.6.3	A.4.3.1
C	**NC**	**N/A**	**U**			
☐	☐	☐	☐	OTHER DIAPHRAGMS: Diaphragms do not consist of a system other than wood, steel deck, concrete, or horizontal bracing.	5.6.5	A.4.7.1
Connections						
C	**NC**	**N/A**	**U**			
☐	☐	☐	☐	STIFFNESS OF WALL ANCHORS: Anchors of concrete or masonry walls to wood structural elements are installed taut and are stiff enough to limit the relative movement between the wall and the diaphragm to no greater than 1/8 in. (3.1 mm) before engagement of the anchors.	5.7.1.2	A.5.1.4

Legend: C = Compliant, NC = Noncompliant, N/A = Not Applicable, U = Unknown

© 2023 American Society of Civil Engineers